In [ ]:

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

customers = pd.read_csv('Ecommerce Customers.csv')

print(customers.head())
print(customers.info())
print(customers.describe())
print(customers.columns)

print(customers.isnull().sum())

sns.pairplot(customers)
plt.show()

sns.histplot(customers['Yearly Amount Spent'])
plt.show()

sns.heatmap(customers.select_dtypes(include='number').corr())
plt.show()

X = customers[['Avg. Session Length', 'Time on App',
               'Time on Website', 'Length of Membership']]
y = customers['Yearly Amount Spent']

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.4, random_state=101
)

from sklearn.linear_model import LinearRegression

lm = LinearRegression()
lm.fit(X_train, y_train)

print(lm.intercept_)

coeff_df = pd.DataFrame(lm.coef_, X.columns, columns=['Coefficient'])
print(coeff_df)

predictions = lm.predict(X_test)

plt.scatter(y_test, predictions)
plt.show()

sns.histplot((y_test - predictions), bins=50)
plt.show()

from sklearn import metrics

print('MAE:', metrics.mean_absolute_error(y_test, predictions))
print('MSE:', metrics.mean_squared_error(y_test, predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test, predictions)))
```

```
                          Email  \
0      mstephenson@fernandez.com
1              hduke@hotmail.com
2               pallen@yahoo.com
3        riverarebecca@gmail.com
4  mstephens@davidson-herman.com

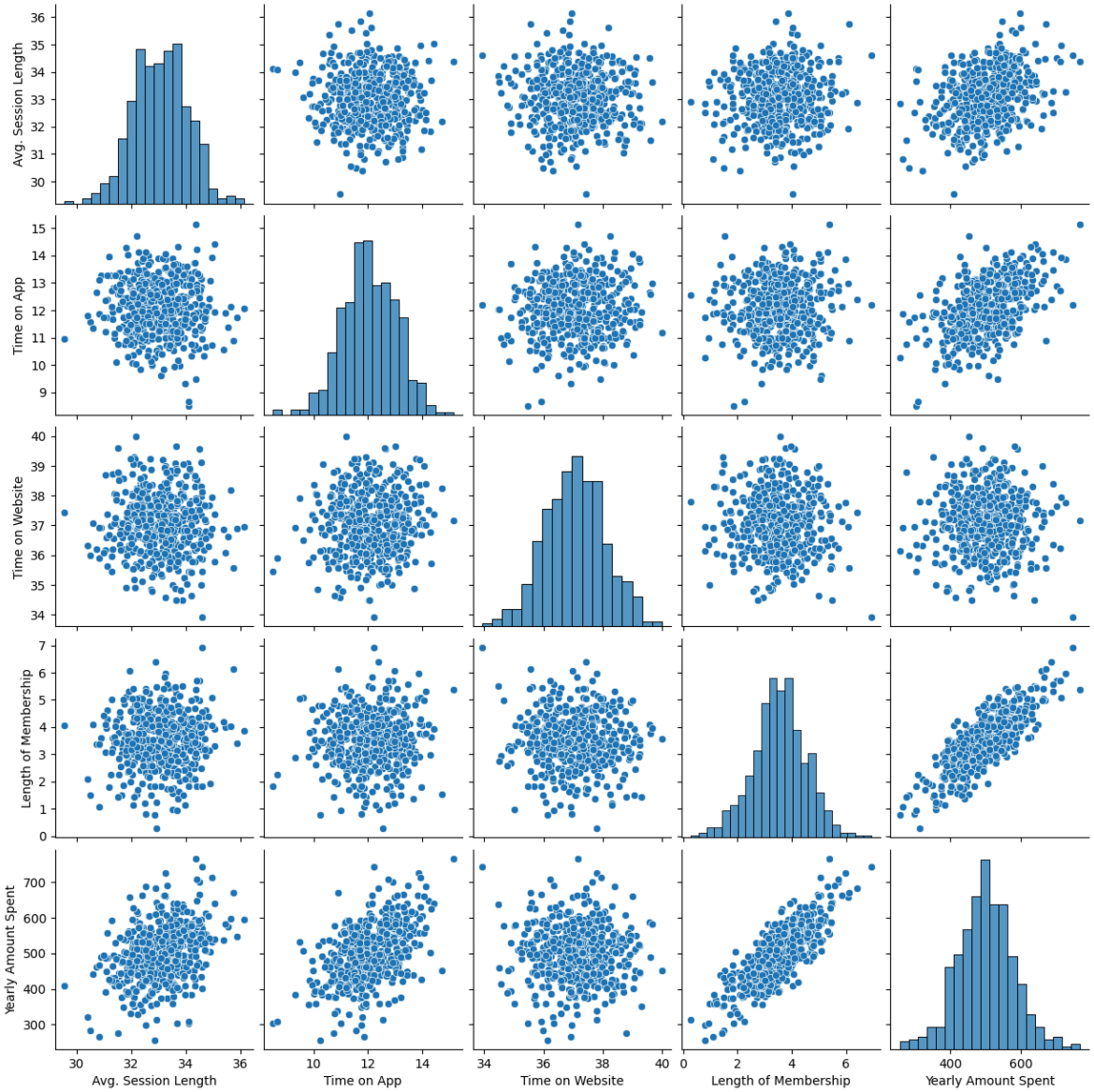
                                        Address            Avatar  \
0       835 Frank Tunnel\nWrightmouth, MI 82180-9605            Violet
1     4547 Archer Common\nDiazchester, CA 06566-8576         DarkGreen
2  24645 Valerie Unions Suite 582\nCobbborough, D...            Bisque
3   1414 David Throughway\nPort Jason, OH 22070-1220       SaddleBrown
4  14023 Rodriguez Passage\nPort Jacobville, PR 3...  MediumAquaMarine

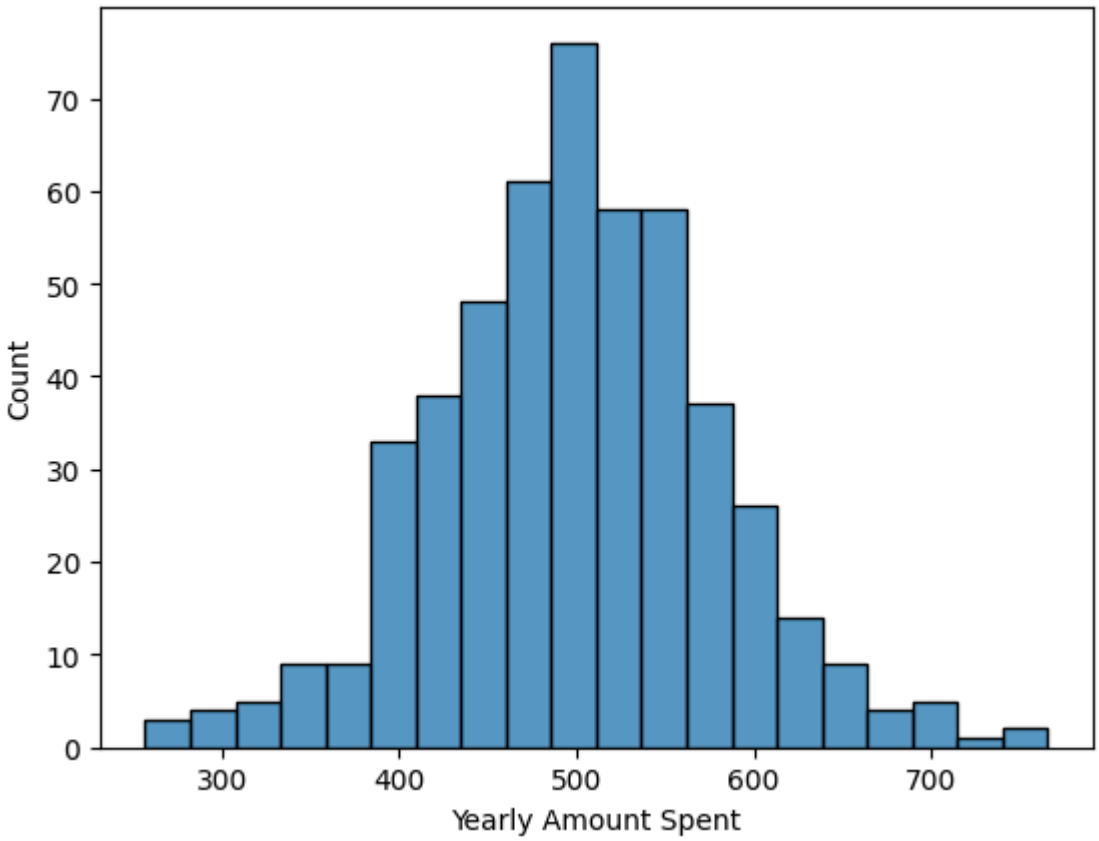
   Avg. Session Length  Time on App  Time on Website  Length of Membership  \
0            34.497268    12.655651        39.577668              4.082621
1            31.926272    11.109461        37.268959              2.664034
2            33.000915    11.330278        37.110597              4.104543
3            34.305557    13.717514        36.721283              3.120179
4            33.330673    12.795189        37.536653              4.446308

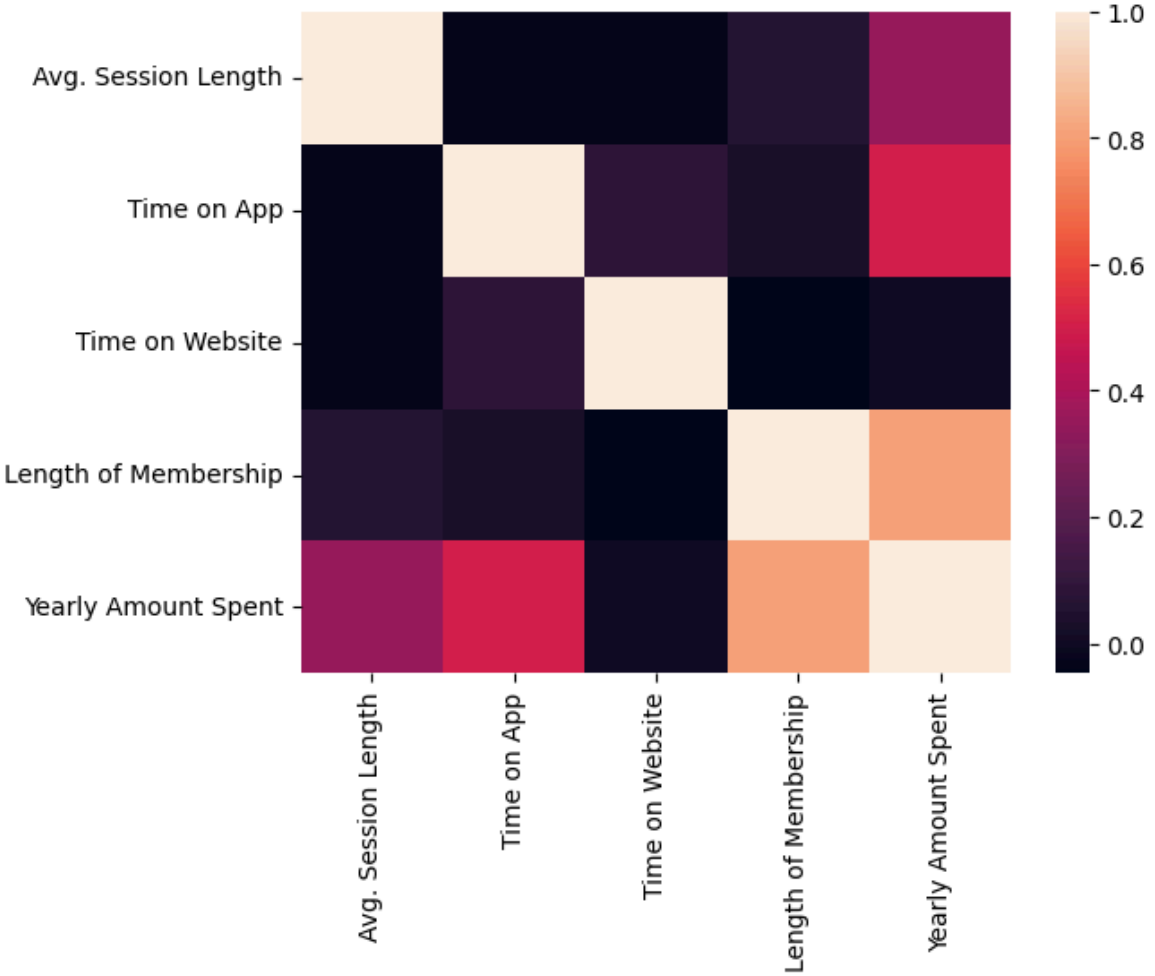
   Yearly Amount Spent
0           587.951054
1           392.204933
2           487.547505
3           581.852344
4           599.406092
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 8 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Email                 500 non-null    object
 1   Address               500 non-null    object
 2   Avatar                500 non-null    object
 3   Avg. Session Length   500 non-null    float64
 4   Time on App           500 non-null    float64
 5   Time on Website       500 non-null    float64
 6   Length of Membership  500 non-null    float64
 7   Yearly Amount Spent   500 non-null    float64
dtypes: float64(5), object(3)
memory usage: 31.4+ KB
None
       Avg. Session Length  Time on App  Time on Website  \
count           500.000000   500.000000       500.000000
mean             33.053194    12.052488        37.060445
std               0.992563     0.994216         1.010489
min              29.532429     8.508152        33.913847
25%              32.341822    11.388153        36.349257
50%              33.082008    11.983231        37.069367
75%              33.711985    12.753850        37.716432
max              36.139662    15.126994        40.005182

       Length of Membership  Yearly Amount Spent
count            500.000000           500.000000
mean               3.533462           499.314038
std                0.999278            79.314782
min                0.269901           256.670582
25%                2.930450           445.038277
50%                3.533975           498.887875
```

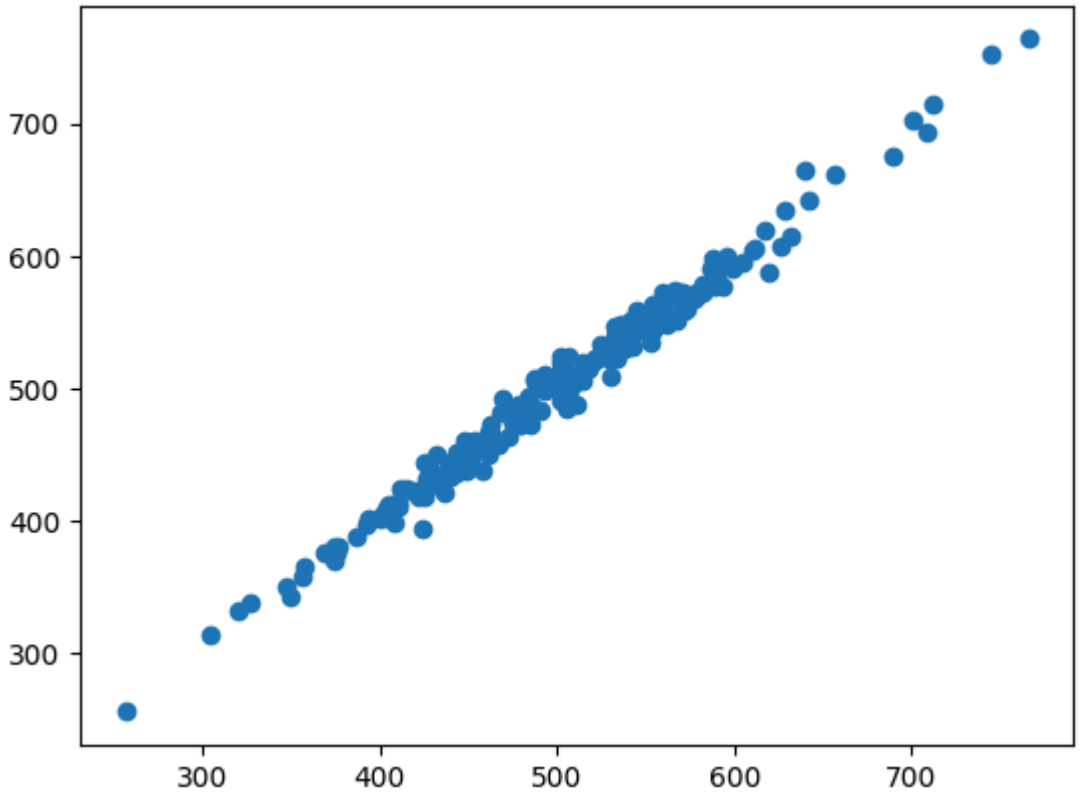
```
75%                  4.126502          549.313828
max                  6.922689          765.518462
Index(['Email', 'Address', 'Avatar', 'Avg. Session Length', 'Time on App',
       'Time on Website', 'Length of Membership', 'Yearly Amount Spent'],
      dtype='object')
Email                   0
Address                 0
Avatar                  0
Avg. Session Length     0
Time on App             0
Time on Website         0
Length of Membership    0
Yearly Amount Spent     0
dtype: int64
```

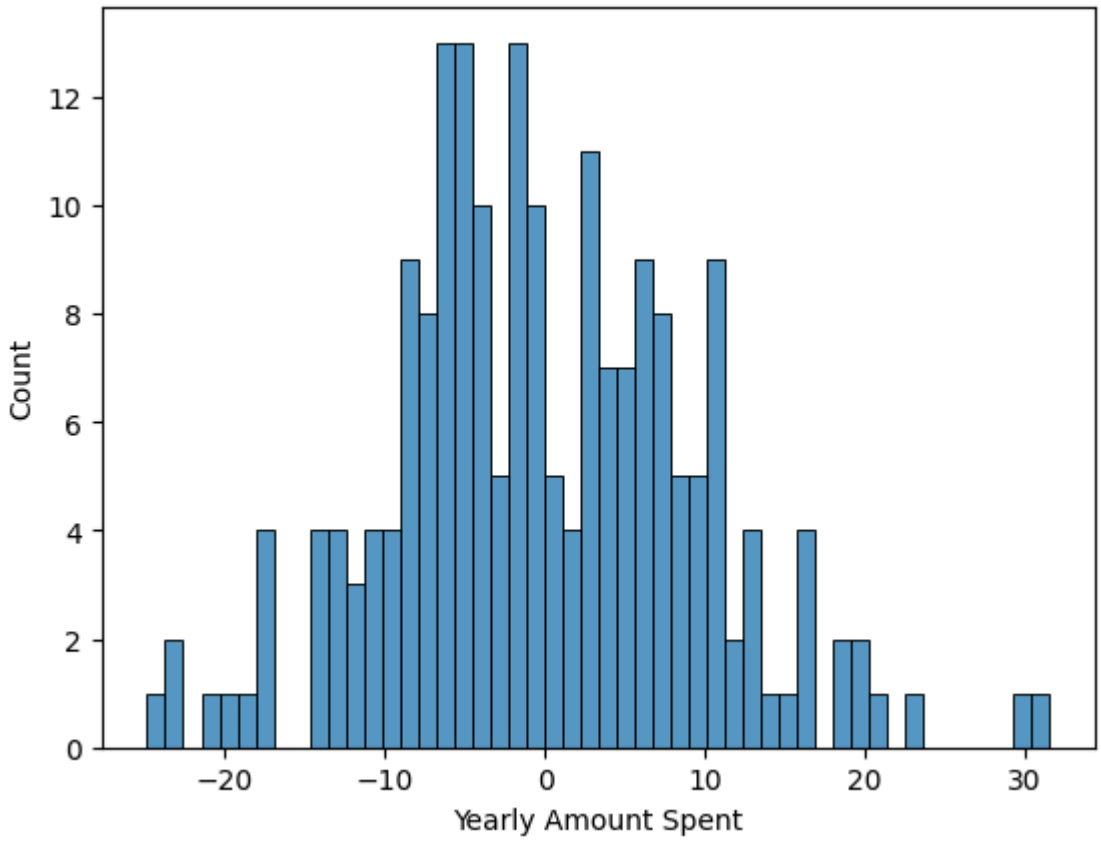



Count
Yearly Amount Spent
0
10
20
30
40
50
60
70
300
400
500
600
700


Avg. Session Length
Time on App
Time on Website
Length of Membership
Yearly Amount Spent
1.0
0.8
0.6
0.4
0.2
0.0

```
-1045.1152168245746
                     Coefficient
Avg. Session Length     25.691540
Time on App             37.892600
Time on Website          0.560581
Length of Membership    61.648594
```






```
MAE: 7.74267128583874
MSE: 93.83297800820075
RMSE: 9.68674238370159
```